

Robotics & Intelligent Control Systems — SHORT NOTES

All Units · Every Formula · Every Trick · Exam Ready

Study *ONLY* these notes → score 100%

Legend: Blue =Circuit Laws Green =Answers/Results Yellow =Motor Violet =Kinematics Teal =Jacobian Red =Danger/Remem

UNIT 7A — CIRCUIT LAWS (Must Know)

Ohm's Law: $V = IR$ $I = V/R$ $R = V/I$

Power: $P = VI = I^2R = V^2/R$ (Watts)

KCL (at any node): $\sum I_{in} = \sum I_{out}$

Sum of all currents = 0. Always 0 around closed loop of voltages.

KVL (around any loop): $\sum V = 0$

Go around loop: battery rise = $\sum$ resistor drops.

Series: $R_{eq} = R_1 + R_2 + \dots$ (same I , add V)

Parallel: $\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$ (same V , add I)

Two resistors: $R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$

Voltage divider: $V_{R1} = V \cdot \frac{R_1}{R_1 + R_2}$

Current divider: $I_{R1} = I \cdot \frac{R_2}{R_1 + R_2}$

RC & RL CIRCUITS (Q9)

RC — Output across C (Low-pass filter):

$$V_{out} = \frac{V_{in}}{1 + j\omega RC}$$

$$|V_{out}| = \frac{V_m}{\sqrt{1 + (\omega RC)^2}} \quad \text{Phase} = -\arctan(\omega RC)$$

Low ω : $V_{out} \approx V_{in}$. High ω : $V_{out} \rightarrow 0$.

RL — Current (Low-pass for current):

$$i = \frac{V_{in}}{R + j\omega L}$$

$$|I| = \frac{V_m}{\sqrt{R^2 + \omega^2 L^2}} \quad \text{Phase} = -\arctan(\omega L/R)$$

Low ω : $i \approx V_{in}/R$. High ω : $i \rightarrow 0$.

Q1–Q6 Quick Answers

Q1: $V_{AB} + V_{BC} + V_{CA} = 0 \text{ V}$ (always! KVL)

Q2: $i = 12/6 = 2 \text{ A}$

Q3: $R_{eq} = 4 + 6 = 10 \Omega$; $i = 10/10 = 1 \text{ A}$

Q4: $i_{12} = 1 \text{ A}$; $i_3 = 10/3 = 3.33 \text{ A}$; Total = 4.33 A

Q5: Drop in direction of i : $\Delta V = -iR = -12 \text{ V}$; magnitude = 12 V

Q6: $R_{12} = 8 \Omega$; $R_{eq} = 8 \parallel 2 = 1.6 \Omega$; $i = 12/1.6 = 7.5 \text{ A}$

MESH ANALYSIS METHOD

Steps:

1. Assign mesh current (clockwise) to each loop.
2. Write KVL for each mesh. Shared resistor: subtract other mesh current.
3. Solve simultaneous equations.

Rule: Traversing resistor with current = drop ($-IR$). Against current = rise ($+IR$). Battery: into + terminal = rise, out of + = drop.

Q7(a): Single mesh, $6 \text{ V} - 2\Omega \cdot i - 2\Omega \cdot i - 4 \text{ V} = 0 \Rightarrow 4i = 2 \Rightarrow i = 0.5 \text{ A}$

Q8(ii): 10 A source sets $I_1 = 10 \text{ A}$. Right mesh: $-20(I_2 - 10) - 30I_2 + 100 = 0 \Rightarrow I_2 = 6 \text{ A}$

UNIT 7B — CAPACITOR & INDUCTOR

Capacitor:

$$i_C = C \frac{dv_C}{dt} \quad v_C = \frac{1}{C} \int i \, dt \quad \text{Energy: } W = \frac{1}{2} C v^2$$

- DC steady state: $dv/dt = 0 \Rightarrow i_C = 0 \Rightarrow$ **open circuit**
- High freq: passes current (low impedance $Z_C = 1/j\omega C$)
- Rising $v_C \Rightarrow +ve \, i_C$ (charging); Falling $\Rightarrow -ve \, i_C$ (discharging)

Inductor:

$$v_L = L \frac{di_L}{dt} \quad i_L = \frac{1}{L} \int v \, dt \quad \text{Energy: } W = \frac{1}{2} L i^2$$

- DC steady state: $di/dt = 0 \Rightarrow v_L = 0 \Rightarrow$ **short circuit**
- Opposes change in current (instantaneously)
- $\frac{di}{dt} \Big|_{t=0+} = \frac{V}{L}$ when $i(0) = 0$

Q10 Formula: $C = 2 \mu\text{F}$; slope of 1 V in $1 \text{ ms} = i_C = 2 \times 10^{-6} \times \frac{1}{10^{-3}} = 2 \text{ mA}$
Rising segment $\rightarrow +2 \text{ mA}$; flat $\rightarrow 0$; falling $\rightarrow -2 \text{ mA}$... BUT slope here is $1 \text{ V}/2 \text{ ms} = 500 \text{ V/s}$, so $i_C = 2\mu \times 500 = 1 \text{ mA}$.

UNIT 6 — DC MOTOR (Most Important!)**The 3 equations — memorise these:**

$$V = IR + K\omega \quad T_e = KI \quad E_b = K\omega$$

General steady-state speed:

$$\omega = \frac{V - IR}{K} = \frac{V - \frac{T_L R}{K}}{K} = \frac{VK - T_L R}{K^2}$$

4 special cases — use these directly:

1. **Const. load** T_L : $I = T_L/K$; $\omega = (V - IR)/K$
2. **No-load** ($T_L = 0$, $I \approx 0$): $\omega_0 = V/K$
3. **Stall** ($\omega = 0$): $I_{\text{stall}} = V/R$; $T_{\text{max}} = KV/R$
4. **Viscous load** $T_L = B\omega$: $\omega = \frac{VK}{BR + K^2}$

Find voltage given ω and T_L : $V = IR + K\omega = \frac{T_L R}{K} + K\omega$ **Real no-load** (friction current I_0 non-zero): $\omega = \frac{V - I_0 R}{K}$ **Post-Midsem Q1–Q12, Q18–Q20 Answers**

Q1	30 rad/s	Q7	26.67 rad/s
Q2	82.5 Nm	Q8	47.5 rad/s
Q3	8.8 A	Q9	12.5 V
Q4	240 rad/s	Q10	16.67 rad/s
Q5	$T_L = 20 \text{ Nm}$; $I = 10 \text{ A}$	Q11	50 rad/s
Q6	96 Nm	Q12	83.33 rad/s
Q18	30.94 rad/s	Q19	26.67 rad/s
Q20	40 rad/s		

H-BRIDGE, PWM, MOSFET, RECTIFIER**PWM:** $V_{dc} = D \cdot V_s$ $D = \frac{t_{on}}{T}$ $T = \frac{1}{f}$ $t_{on} = DT$ **MOSFET regions** (source at GND: $V_{GS} \approx V_G$):

- OFF: $V_{GS} < V_{TH}$
- **Triode (Ohmic):** $V_{DS} < V_{GS} - V_{TH} \leftarrow \text{good switch!}$
- Saturation: $V_{DS} \geq V_{GS} - V_{TH}$

H-Bridge: S1+S4 ON $\rightarrow$ Forward; S2+S3 ON $\rightarrow$ Reverse**Shoot-through** (S1+S2 same leg ON): $\frac{di}{dt}|_{0+} = \frac{V}{L}$ (catastrophic!)**Half-wave rectifier:** $V_{dc} = \frac{V_m}{\pi}$ $I_{dc} = \frac{V_{dc}}{R}$ **Full-wave rectifier:** $V_{dc} = \frac{2V_m}{\pi}$ **Q22:** $D = 30/150 = 0.2$; $t_{on} = 20 \mu\text{s}$; $V_{DS} = 7 < V_{GS} - V_{TH} = 9 \rightarrow$ **Triode****Q23:** $di/dt = 240/(12 \mu) = 20 \text{ MA/s}$ **Q21:** $V_{dc} = 20/\pi = 6.37 \text{ V}$; $I_{dc} = 0.637 \text{ A}$ **MOTOR DYNAMICS — Q24 (Hardest Question)****Full motor equation:** $L \frac{di}{dt} + Ri + E_b = V$ where $E_b = k_e \omega$ **Case 1:** $t \ll L/R$ (ignore R and E_b):

$$L \frac{di}{dt} \approx V \Rightarrow i(t) = \frac{V}{L} t \text{ (ramp)}$$

$$i(1 \text{ ms}) = \frac{120}{0.01} \times 10^{-3} = 12 \text{ A}$$

Valid while $t \ll \tau = L/R = 5 \text{ ms}$ **Electromechanical time constant:**

$$\tau_{eff} = \frac{JL}{k_e k_t} = \frac{0.02 \times 0.01}{0.5^2} = 0.8 \text{ ms}$$

At $t = 5 \text{ ms}$:Pure inductive: $i = 60 \text{ A} \leftarrow \text{wrong!}$ With E_b : $i < 38 \text{ A} \leftarrow \text{realistic}$ **Approximation fails when:** $Ri \gtrsim 0.1V$, i.e. $t \gtrsim 0.1\tau = 0.5 \text{ ms}$. Back-EMF builds as ω grows, opposing the current rise.**UNIT 1 — ROTATION MATRICES****3 basic rotations:**

$$R_x(\phi) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & c_\phi & -s_\phi \\ 0 & s_\phi & c_\phi \end{pmatrix} \quad R_y(\theta) = \begin{pmatrix} c_\theta & 0 & s_\theta \\ 0 & 1 & 0 \\ -s_\theta & 0 & c_\theta \end{pmatrix}$$

$$R_z(\psi) = \begin{pmatrix} c_\psi & -s_\psi & 0 \\ s_\psi & c_\psi & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

KEY properties (always tested):

$$R^{-1} = R^T \quad \det(R) = +1 \quad R \in SO(3)$$

- Columns = unit vectors, mutually orthogonal
- Rows also orthonormal
- Non-commutative: $R_x R_z \neq R_z R_x$

YPR: $R = R_z(\psi)R_y(\theta)R_x(\phi)$ (fixed axes, post-multiply)**Body frame:** post-multiply; **World frame:** pre-multiply**Key angles:**

θ	0°	30°	45°	$60^\circ/90^\circ$
sin	0	0.5	0.707	0.866/1
cos	1	0.866	0.707	0.5/0

UNIT 1 — HTM (4×4 Matrix)**Structure:**

$$T = \begin{pmatrix} R_{3 \times 3} & \vec{p}_{3 \times 1} \\ \mathbf{0}_{1 \times 3} & 1 \end{pmatrix}$$

 R = rotation (orientation), $\vec{p}$ = position of frame origin**Inverse (no matrix inversion needed!):**

$$T^{-1} = \begin{pmatrix} R^T & -R^T \vec{p} \\ \mathbf{0} & 1 \end{pmatrix}$$

Composition: ${}^0T_n = {}^0T_1 \cdot {}^1T_2 \cdots {}^{n-1}T_n$ **Point transform:** ${}^A p = T \cdot {}^B p$ (append 1 to make 4×1)**Q15 template:** $R_z(90^\circ)$, translate $(4, 3, 0)$:

$${}^A T_B = \begin{pmatrix} 0 & -1 & 0 & 4 \\ 1 & 0 & 0 & 3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Q14 trick: Translate then rotate — the translation vector also gets rotated!

$$(3, 0, 0) \xrightarrow{R_z(90^\circ)} (0, 3, 0) \rightarrow \text{answer: } [0, 3, 0]^T$$

UNIT 2 — DH PARAMETERS & FK**4 parameters per joint:**

Param	Meaning	Variable?
θ_i	Angle about Z_{i-1}	YES (revolute)
d_i	Offset along Z_{i-1}	YES (prismatic)
a_i	Length along X_i	No (fixed)

UNIT 3 — INVERSE KINEMATICS

IK problem: Given desired (x, y) , find (θ_1, θ_2) .

2-DOF IK formulas (MEMORISE):

$$\cos \theta_2 = \frac{x^2 + y^2 - L_1^2 - L_2^2}{2L_1 L_2}$$

$\theta_2 = \pm \arccos(\dots)$ (two solutions: elbow up/down)

$$k_1 = L_1 + L_2 \cos \theta_2, \quad k_2 = L_2 \sin \theta_2$$

$$\theta_1 = \text{atan2}(y, x) - \text{atan2}(k_2, k_1)$$

Workspace check: Point reachable iff $|L_1 - L_2| \leq r \leq L_1 + L_2$

Q25: $(5, 5)$: $r = 7.07 > L_1 + L_2 = 7 \rightarrow$ **NO solution**

Key facts:

- Outside workspace $\rightarrow$ **no solution** (Q29: b)
- 3R robot $\rightarrow$ typically **2 solutions** (Q28: b)
- Closed-form exists when **3 adjacent axes intersect/parallel** (Q30: a)
- Analytical=fast, exact; Numerical=iterative, always works

UNIT 4 — JACOBIAN & SINGULARITIES

Definition: $\dot{\vec{x}} = J(\vec{q})\dot{\vec{q}}$ maps joint velocity $\rightarrow$ task velocity

2-DOF Jacobian (derive by differentiating FK):

$$J = \begin{pmatrix} -L_1 s_1 - L_2 s_{12} & -L_2 s_{12} \\ L_1 c_1 + L_2 c_{12} & L_2 c_{12} \end{pmatrix}$$

Key determinant: $\det J = L_1 L_2 \sin \theta_2$

Dexterity: $w = |\det J| = L_1 L_2 |\sin \theta_2|$

Singularities ($\det J = 0, w = 0$): $\theta_2 = 0^\circ$ or 180°

$\theta_2 = 0^\circ$: arm fully extended (boundary)

$\theta_2 = 180^\circ$: arm folded back (boundary)

Effect: Can't move in some direction; joint vel $\rightarrow \infty$

1R arm Jacobian: $J = (-l \sin \theta, l \cos \theta)^T$

At $\theta = 0^\circ$: $J = (0, l)^T = [0, 0.8]^T$ (Q31: a)

Never singular (1-DOF, always rank 1) (Q32: c)

At $\theta = 90^\circ$: $\dot{x} = -l\dot{\theta}$; for $\dot{x} = 0.4 \Rightarrow \dot{\theta} = -0.5$ (Q33: d)

7-DOF Jacobian is 6×7 : non-square, no exact inverse (Q34: a)

Use pseudoinverse: $J^\dagger = J^T (J J^T)^{-1}$

UNIT 5 — FORCE, STATICS & FRAME ROTATION

Force-torque relation:

$$\vec{\tau} = J^T \vec{F}$$

- $\vec{\tau}$: joint torques vector
- $\vec{F}$: end-effector wrench (forces + moments)
- J^T : transpose of Jacobian

Virtual work proof: $\vec{\tau}^T \delta \vec{q} = \vec{F}^T \delta \vec{x}$ and $\delta \vec{x} = J \delta \vec{q}$

$\Rightarrow \vec{\tau}^T = \vec{F}^T J \Rightarrow \vec{\tau} = J^T \vec{F}$ ✓

At singularity: Small end-effector force $\rightarrow$ huge joint torques.

Frame rotation facts (Q13):

- R_{sc} = columns are $\hat{x}_c, \hat{y}_c, \hat{z}_c$ in $\{s\}$ coords
- $\hat{z} = \hat{x} \times \hat{y}$ (right-hand rule)
- Post-multiply (RR_2): rotate about **body** axis
- Pre-multiply (R_2R): rotate about **world** axis
- $R_{sb} p_b$: change of reference frame
- $R_{sb} p_s$: moves the point

EXAM TRICKS & THINGS THAT ALWAYS APPEAR

Circuit tricks:

- $V_{AB} + V_{BC} + V_{CA} = 0$ always (KVL loop)
- DC capacitor = open circuit ($i = 0$)
- DC inductor = short circuit ($v = 0$)
- Capacitor current square wave when voltage is triangular
- Power check: $\sum P_{delivered} = \sum P_{absorbed}$ (always equal)

Motor tricks:

- Always start: find $I = T_L / K$ first, then ω
- Viscous load formula: $\omega = VK / (BR + K^2)$
- Voltage drop $\Delta V / K = \Delta \omega$ (for const. T_L)
- Real no-load: $\omega = (V - I_0 R) / K$ not V / K

Kinematics tricks:

- $R^{-1} = R^T$ (never invert by row reduction!)
- $\det(R) = +1$ always
- HTM inverse: $-R^T \vec{p}$ in translation part
- Planar robot: $\alpha_i = 0$ and $d_i = 0$
- θ is variable for revolute joint
- Check reachability before IK: $r \leq L_1 + L_2$
- Singularity: $\theta_2 = 0^\circ$ or 180° (when $\det J = 0$)
- 4n DH parameters for n-axis robot

Quick MCQ answers:

Q7(K): A^T Q8(K): $(0, 1, 0)$ Q9(K): $\det = +1$

Q14: $[0, 3, 0]^T$ Q16: θ Q26: 4n

Q27: $(0.4, 0.4\sqrt{3})$ Q28: 2 Q29: No

Q30: (a) Q31: $[0, 0.8]$ Q32: Never Q33: -0.5 Q34: (a)

HTM Point Transform (Q18):

$$\begin{aligned} A_{TB} \cdot (1, 2, 3, 1)^T: \\ x &= 0(1) - 1(2) + 0 + 3 = 1 \\ y &= 1(1) + 0 + 0 + 5 = 6 \\ z &= 0 + 0 + 3 + 2 = 5 \\ \Rightarrow (1, 6, 5) \end{aligned}$$

HTM Inverse (Q19):

$$\begin{aligned} R^T &= \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ -R^T d &= (-6, 4, -3)^T \\ \Rightarrow T^{-1} &= \begin{pmatrix} 0 & 1 & 0 & -6 \\ -1 & 0 & 0 & 4 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 1 \end{pmatrix} \end{aligned}$$

Jacobian at $\theta_1 = 0, \theta_2 = 90^\circ, L = 0.5$:

$$\begin{aligned} s_1 = 0, c_1 = 1, s_{12} = 1, c_{12} = 0 \\ J &= \begin{pmatrix} -0.5 & -0.5 \\ 0.5 & 0 \end{pmatrix} \\ \det J &= 0.25 \quad w = 0.25 \\ \dot{\vec{x}} &= J(2, 1)^T = (-1.5, 1)^T \end{aligned}$$

2-link FK (Q22):

$$\begin{aligned} x &= 4 \cos 30 + 3 \cos 75 \\ &= 3.464 + 0.776 = \mathbf{4.24} \\ y &= 4 \sin 30 + 3 \sin 75 \\ &= 2 + 2.898 = \mathbf{4.90} \\ \text{End-effector: } &(4.24, 4.90) \end{aligned}$$

DH Matrix (Q21):

$$\begin{aligned} \theta &= 45^\circ, a = 5, d = 0, \alpha = 0: \\ T &= \begin{pmatrix} 0.707 & -0.707 & 0 & 3.536 \\ 0.707 & 0.707 & 0 & 3.536 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \end{aligned}$$

Tool tip (Q23):

$${}^0T_E \cdot (1, 0, 2, 1)^T:$$

$$x = 0.866 + 0 + 0 + 8 = 8.866$$

$$y = 0.5 + 0 + 0 + 4 = 4.5$$

$$z = 0 + 0 + 2 + 6 = 8$$

$$\Rightarrow (8.866, 4.5, 8)$$

MOTOR FORMULA DECISION TREE — Use this in exam: